Alberto Clocks

Design Document

Instructor.	Sir M. Arqum Tariq	
Batch.	2606G2	
Group.	D	
Serial No.	**Enrollment Number**	**Student Name**
1.	1732077	Fatima Raza
2.	1732701	Alishah Sabri
3.	1733431	Areeba Sabri

Aptech eProject

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Acknowledgement

First, we would like to thank our instructor, Sir M. Arqum Tariq, for his guidance and support during this project. His suggestions helped us improve our understanding of website planning, design and development.

We are also thankful to the eProject team at Aptech Head Office for giving us the opportunity to work on Alberto Clocks. This project helped us apply the HTML, CSS and JavaScript skills learned during our course.

Finally, we thank everyone who helped us review the website, test its features and prepare the project documents.

- The project gave us practical experience in web development.
- It helped us understand how to divide a website into sections.
- It improved our skills in teamwork, testing and documentation.

Synopsis

Alberto Clocks is a website made to present wall, luxury and vintage clocks. It helps visitors view the collection, compare clocks and read information about the store.

The website contains a Home section, product collection, product details page, shopping cart, wishlist, login and registration forms, reviews, store information and a Support section.

HTML is used for page structure, CSS for design, JavaScript for interactive features and a JSON file for the product list. Browser storage keeps demonstration information such as cart items, wishlist, account details, reviews and orders.

Checkout demonstrates the order process but does not take real payment. Order status is also a demonstration. The project is intended for learning front-end web development.

Problem Definition

Customers need a simple way to explore clocks before visiting a showroom. A website should show product pictures, prices and features. Visitors should also be able to find a clock without checking every product manually.

The project aims to answer these questions:

- How should the Home page and Product section be arranged?
- How should clocks be divided into Wall, Luxury and Vintage categories?
- How can visitors search and sort the collection?
- What information should appear on a product card and detail page?
- How should the cart and checkout screens be arranged?
- How should login and registration forms guide the visitor?
- How can the website fit desktop, tablet and mobile screens?
- How should the store address, map and support information be shown?
- What should happen when product data or images fail to load?

Customer Requirement Specification (CRS)

The main requirements for Alberto Clocks are:

1. The Home section must show the Alberto Clocks name and the slogan *Time is Luxury*.
2. The header must have Home, Product, Technology, Store Locator, Support and Register links.
3. The Home section must show a digital clock, date, store status and LED colour buttons.
4. Products must be grouped into Wall, Luxury and Vintage categories.
5. Visitors must be able to search, sort and filter products by price.
6. Product cards must show pictures, names, categories and prices.
7. The details page must show a larger image, description, specifications and stock information.
8. Visitors must be able to add products to the cart and change quantities.
9. The website must provide a wishlist and product reviews.
10. Registration must check password rules and password confirmation.
11. Checkout must display customer details, order items and total.
12. The Store Locator must show the address, opening hours and map.
13. The Support section must contain FAQs and a contact form.
14. The layout must fit desktop and mobile screens.
15. The website must show useful messages when data or images fail to load.

Scope of the Work

The project covers the front-end design and development of Alberto Clocks using HTML, CSS, JavaScript, Bootstrap and JSON.

The work includes the Home page, product collection, search and filters, product details, cart, wishlist, account forms, reviews, checkout demonstration, order history, Store Locator and Support section. It also includes the digital clock, date, ticker, world-time cards, theme controls and mobile navigation.

The current version is a learning project. Accounts, cart items, wishlist, reviews, messages and orders are saved in the current browser. Checkout does not take payment and the website does not send real orders or emails.

1. System Requirements

1.1 Software Requirements

- Operating system: Windows, macOS or Linux.
- Code editor: Visual Studio Code or another text editor.
- Browser: current Chrome, Edge, Firefox or Safari.
- Local server: VS Code Live Server or another HTTP server.
- Technologies: HTML5, CSS3, JavaScript, Bootstrap and JSON.
- Internet connection for remote photographs, map content and the configured checkout connection check.

1.2 Hardware Requirements

A standard computer or laptop that can run a current browser and code editor is sufficient. A phone or tablet can be used to check the mobile layout.

1.3 How to Open the Website

1. Extract the E-project folder.
2. Open the folder in Visual Studio Code.
3. Install Live Server if needed.
4. Right-click `index.html`.
5. Select **Open with Live Server**.

A local server is needed because the website loads product information from `data/products.json`. If the list is empty, check the file path and the browser console.

1.4 Project Limitations

- Some product pictures and the map need an Internet connection.
- Account, cart, wishlist, reviews and order information stays in the current browser.
- Saved information is not shared with other devices.
- Checkout does not take real payment.
- Order status is simulated.
- Contact and newsletter forms do not send real emails.

2. Project Folder Structure

The project is divided into folders for HTML pages, CSS styles, JavaScript functions, product data, images and fonts.

```
E-project/
|-- index.html
|-- README.md
|
|-- css/
|   |-- about.css
|   |-- accessibility.css
|   |-- auth.css
|   |-- base.css
|   |-- bootstrap.min.css
|   |-- cart-footer.css
|   |-- features.css
|   |-- forms-faq.css
|   |-- header.css
|   |-- hero-bg.css
|   |-- hero.css
|   |-- led.css
|   |-- pages.css
|   |-- product-photos.css
|   |-- products.css
|   |-- responsive.css
|   |-- sections.css
|   |-- store-status.css
|   |-- style.css
|   |-- theme-toggle.css
|
|-- data/
|   |-- products.json
|
|-- fonts/
|   |-- DSEG7Classic-Regular.woff2
|   |-- README.txt
|
|-- image/
|   |-- about-mechanism.svg
|   |-- about-wall-clock.svg
|   |-- favicon.svg
|   |-- hero-bg.svg
```

```
|   |-- register-dial.svg
|   |-- products/
|       |-- clock-1.svg ... clock-24.svg
|
|-- js/
|   |-- auth.js
|   |-- bootstrap.bundle.min.js
|   |-- cart.js
|   |-- core.js
|   |-- forms.js
|   |-- main.js
|   |-- orders.js
|   |-- products.js
|   |-- reviews.js
|   |-- security.js
|   |-- ui.js
|   |-- wishlist.js
|
|-- pages/
    |-- product.html
```

2.1 HTML Pages

`index.html` contains the main website sections and the interface elements for login, cart, wishlist, checkout and profile.

`pages/product.html` displays the full details of a selected clock. The product ID is passed in the page address.

2.2 CSS Files

The file `css/style.css` imports the other CSS files in order. The stylesheets control the header, hero, product cards, forms, cart, themes, accessibility and responsive layout.

2.3 JavaScript Files

The files inside `js/` control product loading, cart operations, accounts, wishlist, reviews, forms and interface actions.

2.4 Data, Images and Fonts

The file `data/products.json` holds the product details used by the website. The folder `image/` contains the illustrations and product pictures. The folder `fonts/` contains the digital clock font and instructions for optional fonts.

3. Aggregated Sitemap

The sitemap shows the main website sections and the links to product details, cart, account and support features.

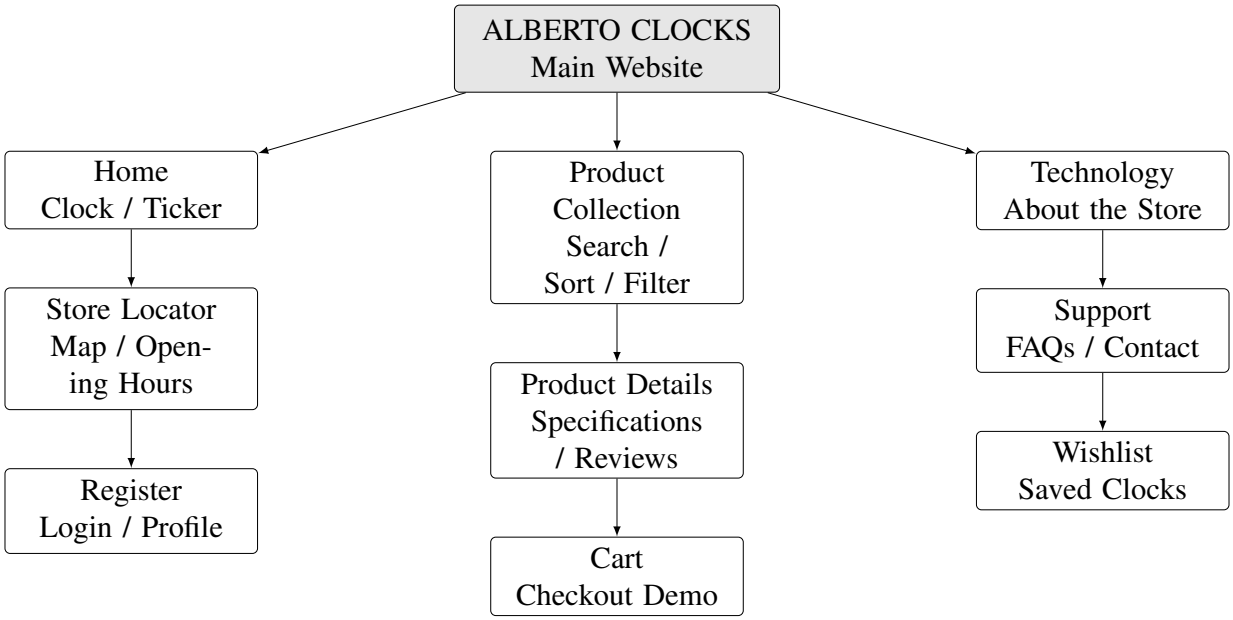


Figure 1: *Aggregated sitemap of Alberto Clocks*

4. Wireframes

The wireframes show where the main elements are placed. They are simple black-and-white drawings of the website screens. They are not screenshots of the finished website.

4.1 Main Website Layout

The main page contains the header, navigation, ticker, Home section, Product section, Technology section, Store Locator, Support and footer.



Figure 2: *Wireframe: Main website layout*

4.2 Home Section

The Home section focuses on the digital clock, store status and collection link. World-time cards appear below the main clock area.



Figure 3: *Wireframe: Home section*

4.3 Products Section

The Products section contains search and sort controls, category buttons, product cards and page controls. Each card shows a clock image, name, price and Add to Cart button.

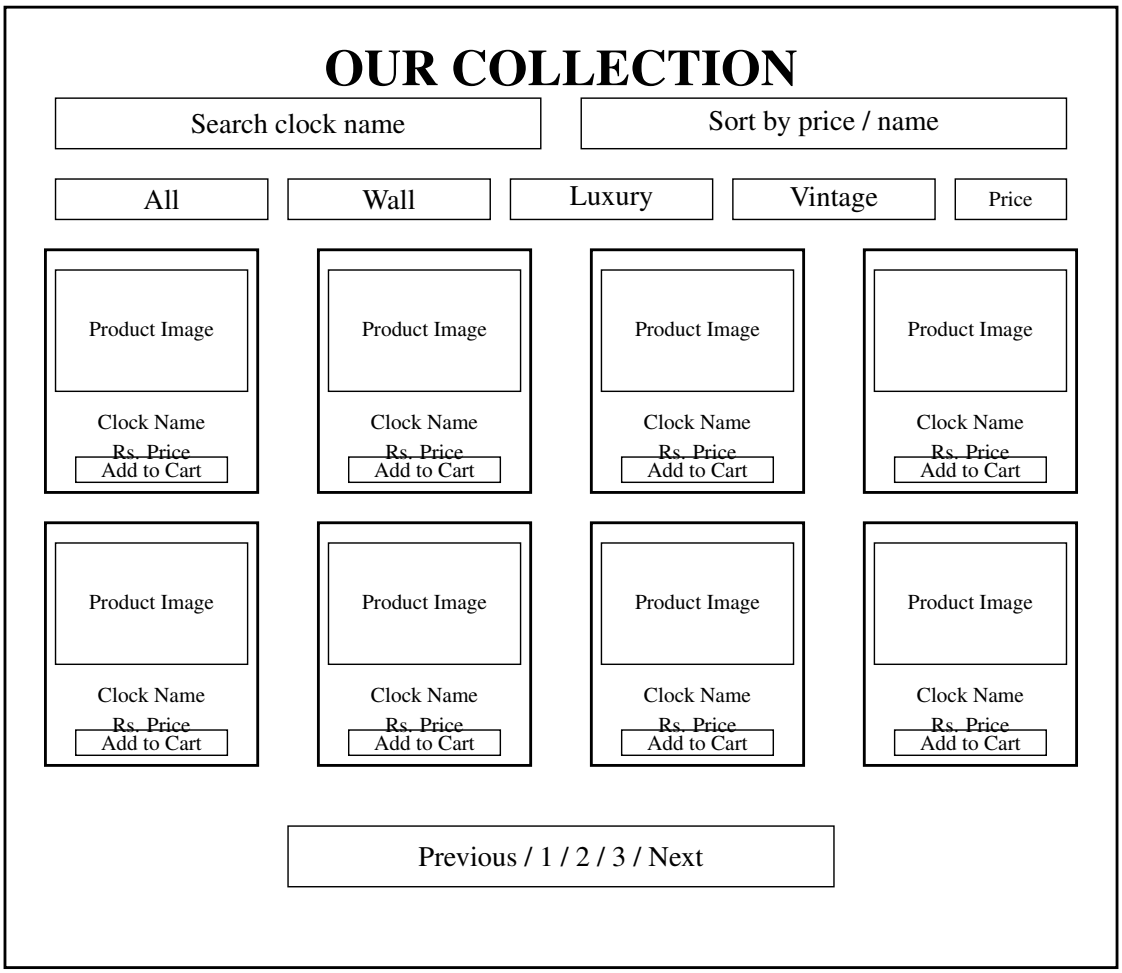


Figure 4: *Wireframe: Products section*

4.4 Product Details Popup

The popup gives a quick view of a clock without leaving the collection.

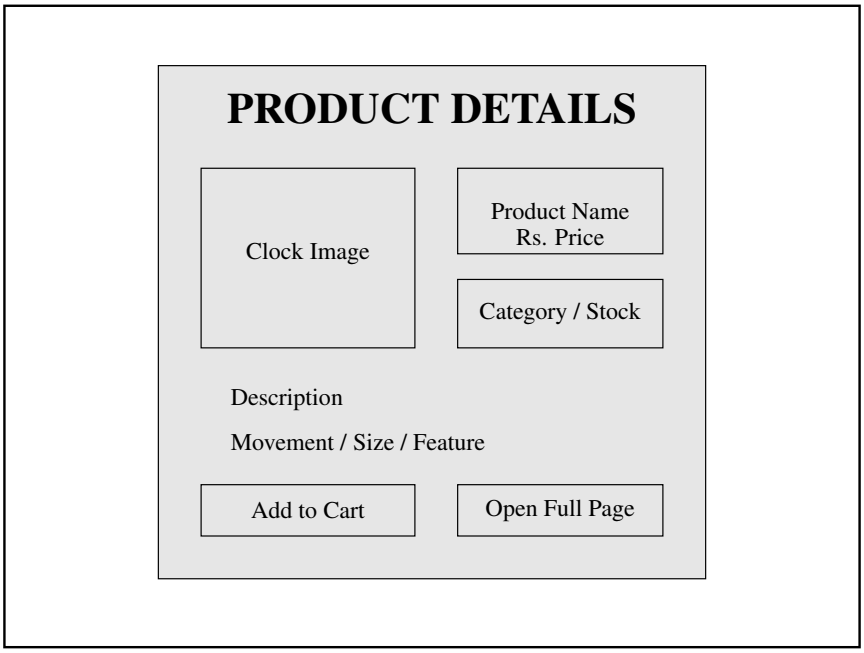


Figure 5: *Wireframe: Product details popup*

4.5 Product Details Page

The detail page shows a large picture, specifications, stock, quantity controls, wishlist, related clocks and reviews.

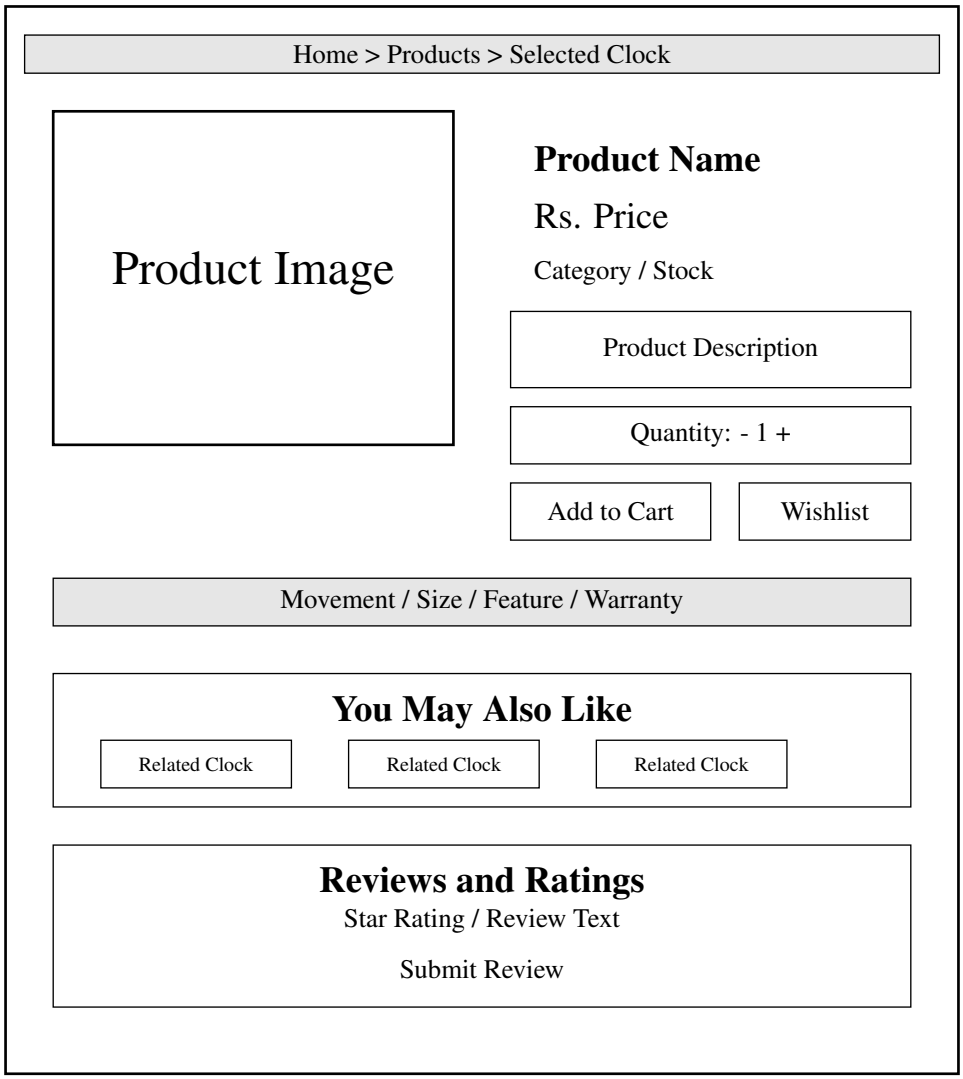


Figure 6: *Wireframe: Product details page*

4.6 Product Line-Up

Wall, Luxury and Vintage are categories within the same product collection. The category buttons change which clocks appear in the grid.

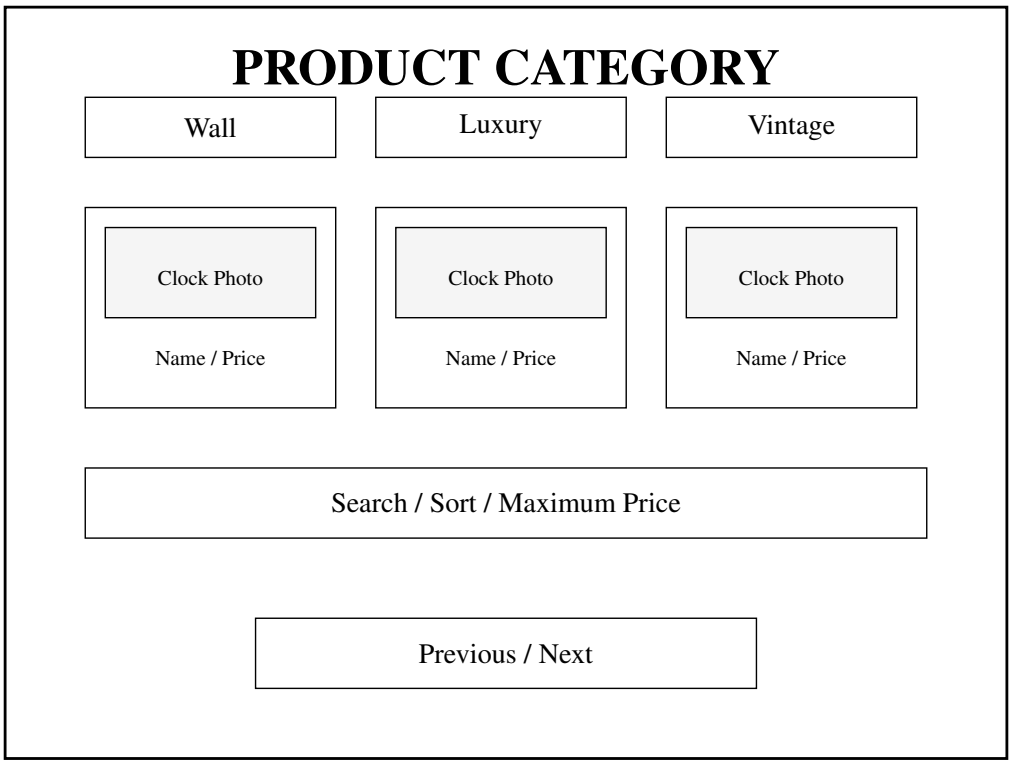


Figure 7: *Wireframe: Product categories*

4.7 Technology Section

The Technology section provides information about Alberto Clocks and features of the clock collection.

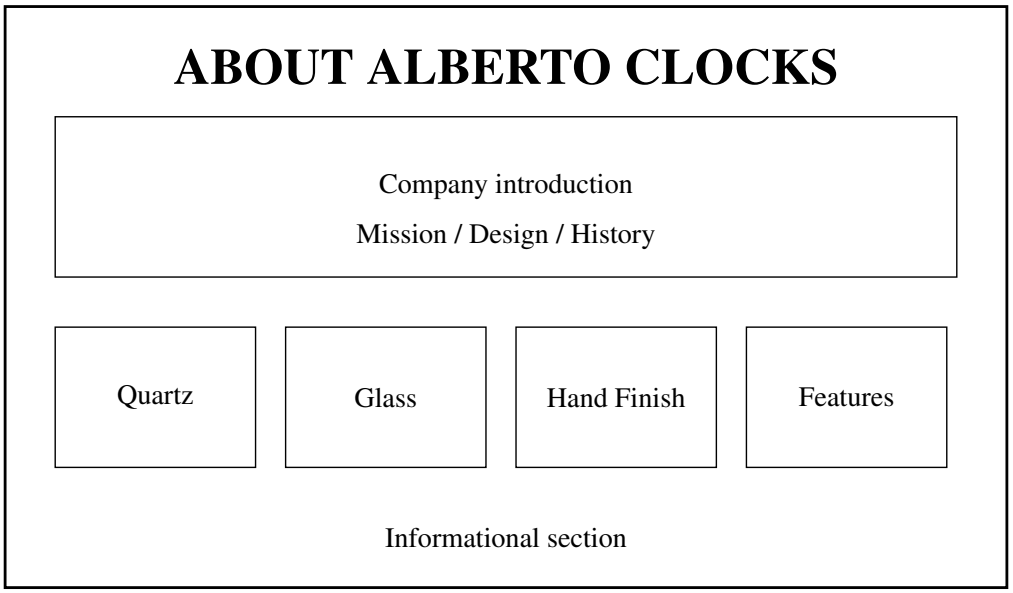


Figure 8: *Wireframe: Technology section*

4.8 Store Locator Section

The map is placed beside the showroom details and opening hours.

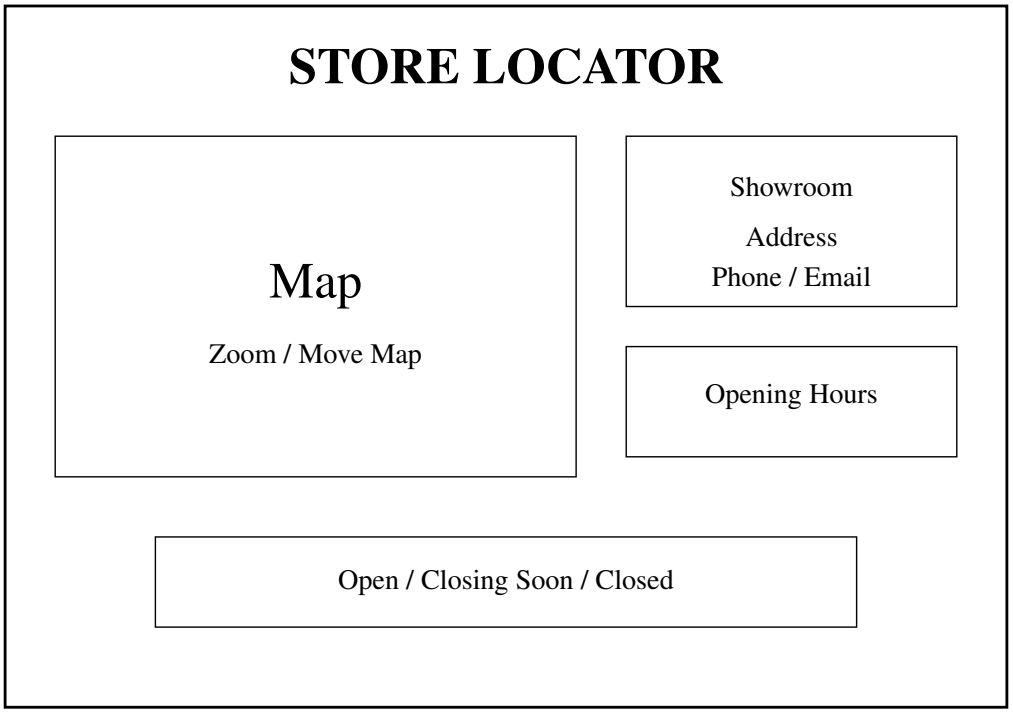


Figure 9: *Wireframe: Store Locator section*

4.9 Support Section

The Support section contains contact information, FAQs and a form for the visitor's name, email and message.

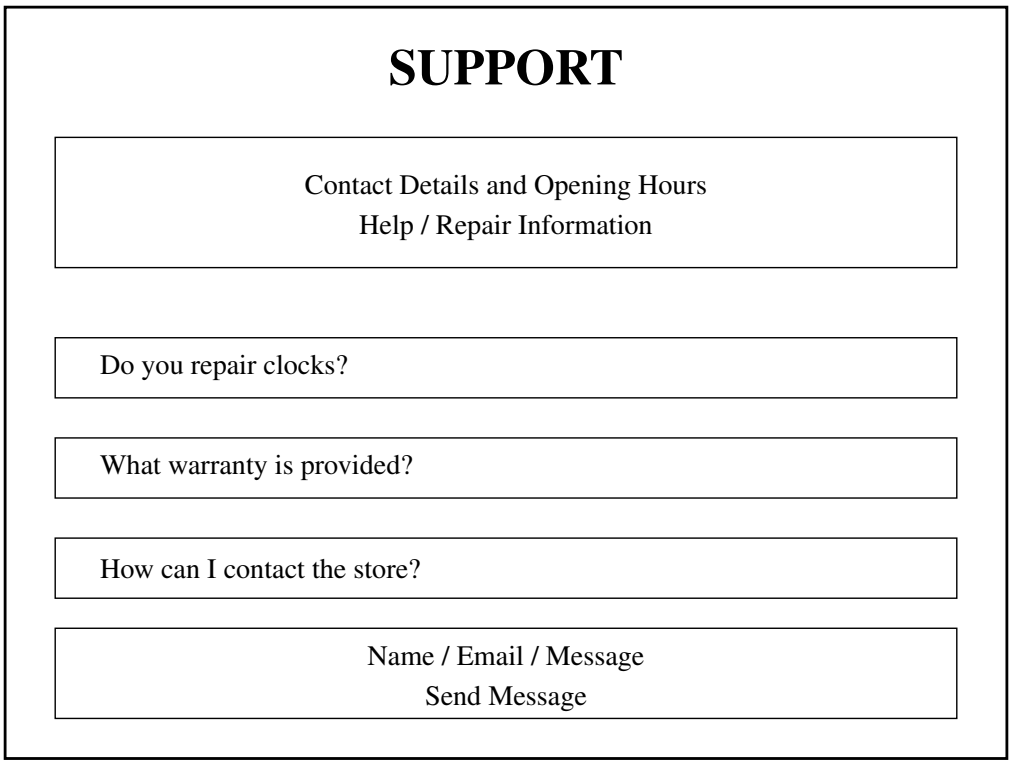


Figure 10: *Wireframe: Support and FAQ section*

4.10 Register Section

The Register form contains the fields required to create a demonstration account.

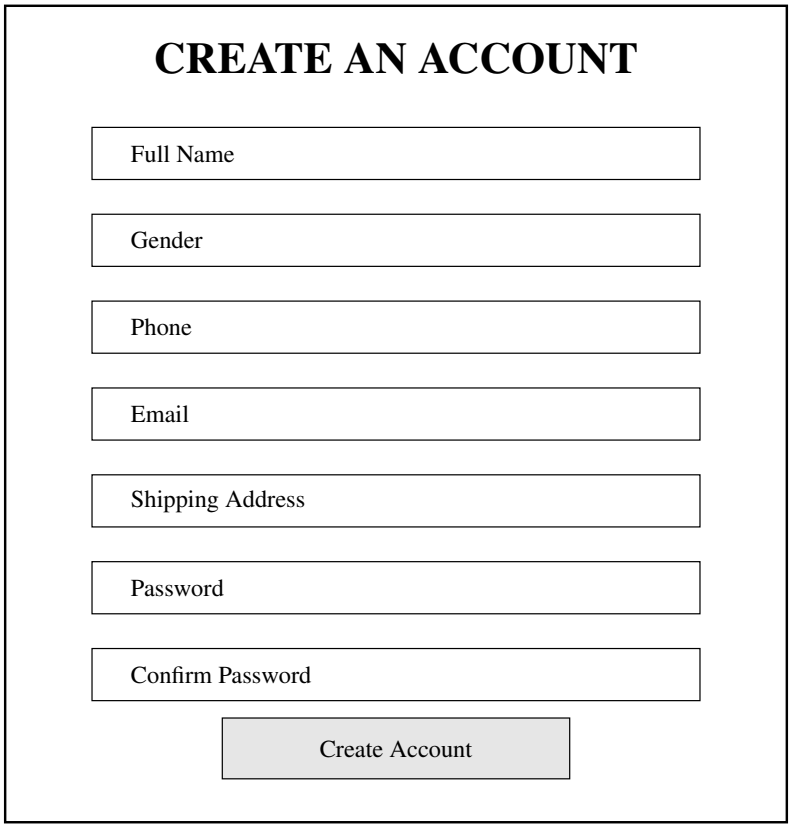


Figure 11: *Wireframe: Register section*

4.11 Footer

The footer displays site links, store information, opening hours and the newsletter form.

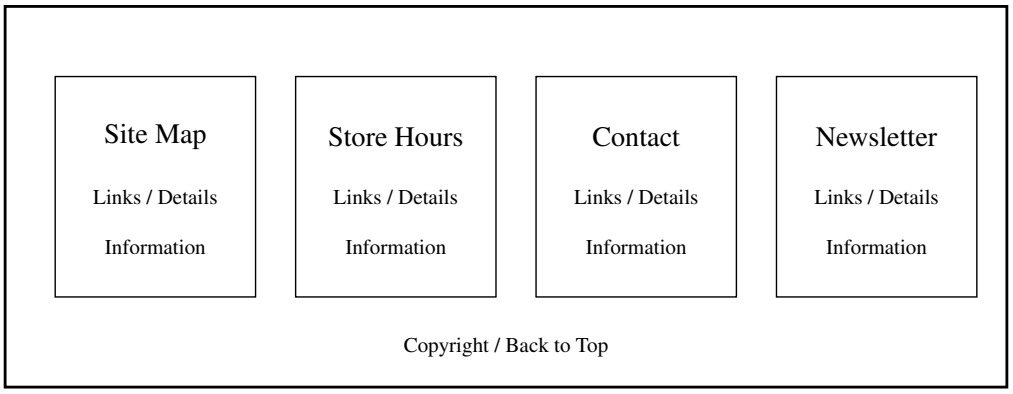


Figure 12: *Wireframe: Footer*

4.12 Login and Sign-Up Window

The login window contains Login and Sign Up tabs. Registration has extra fields and password rules.



Figure 13: *Wireframe: Login and Sign-Up window*

4.13 Cart Panel

The cart panel contains the selected clocks, quantity controls, coupon field, subtotal, discount and total.

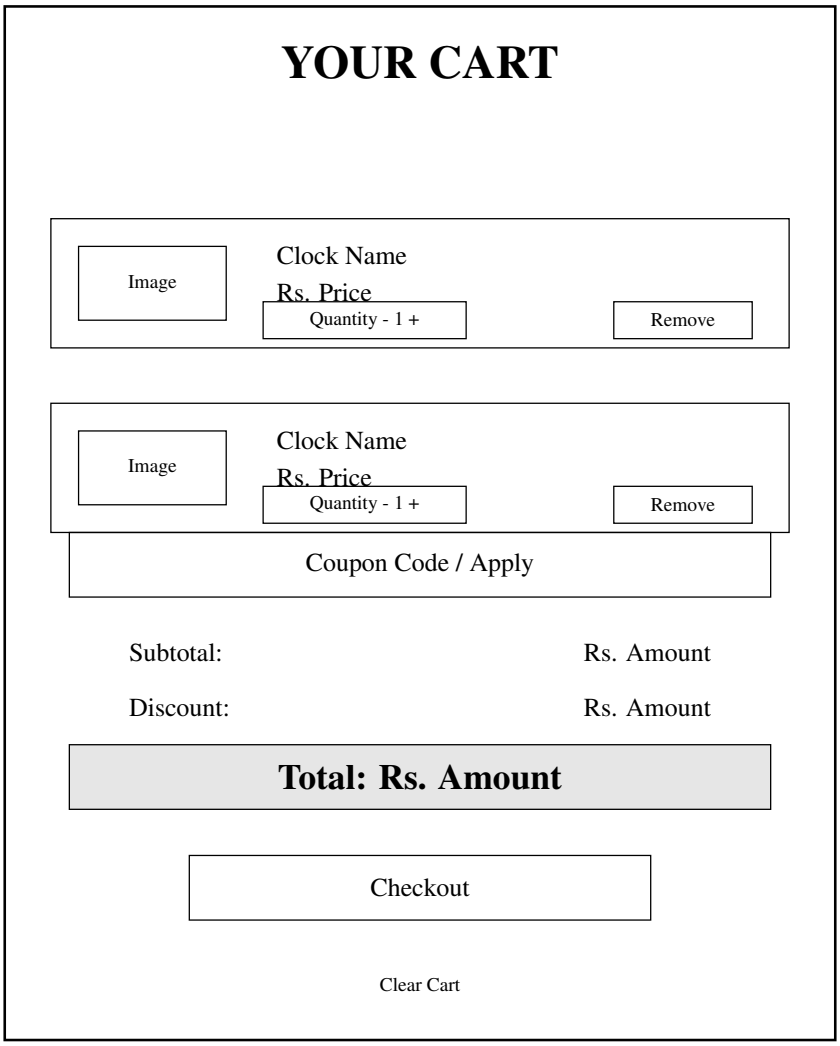


Figure 14: *Wireframe: Cart panel*

4.14 Profile Window

The profile displays account details and saved demonstration orders.

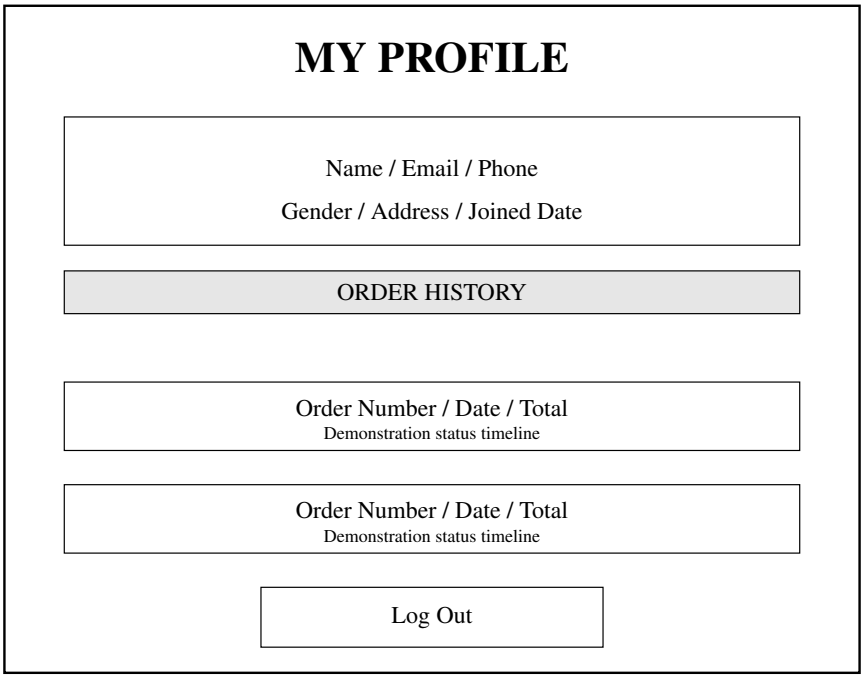


Figure 15: *Wireframe: Profile and order history*

4.15 Mobile Layout

On a narrow screen, the menu is collapsed and product cards use a smaller layout. Controls fit inside the screen.

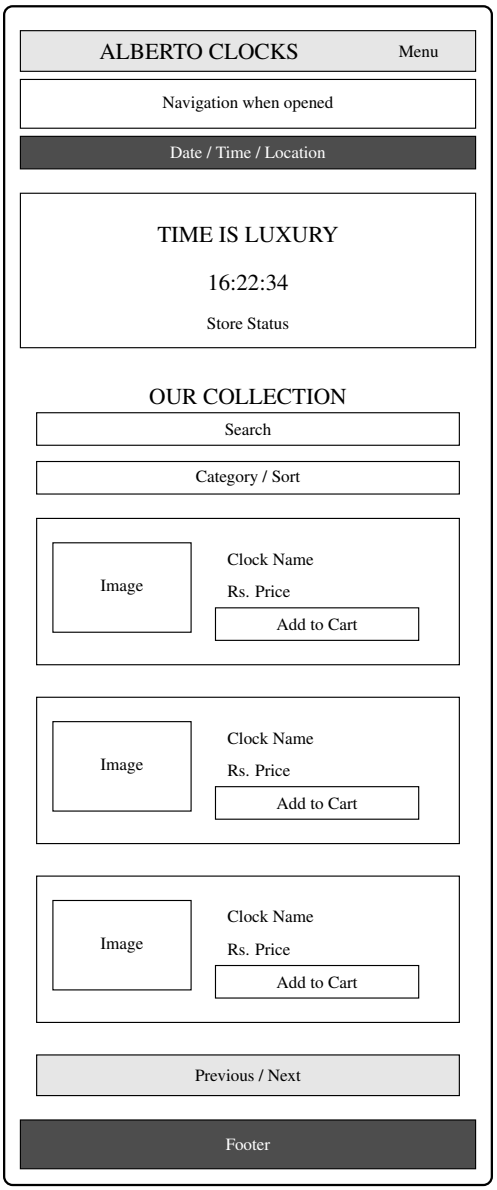


Figure 16: *Wireframe: Mobile layout*

4.16 Wishlist Panel

The wishlist panel shows clocks saved using the heart button.



Figure 17: *Wireframe: Wishlist panel*

4.17 Checkout Dialog

The checkout dialog collects customer details and shows the order summary before submission.

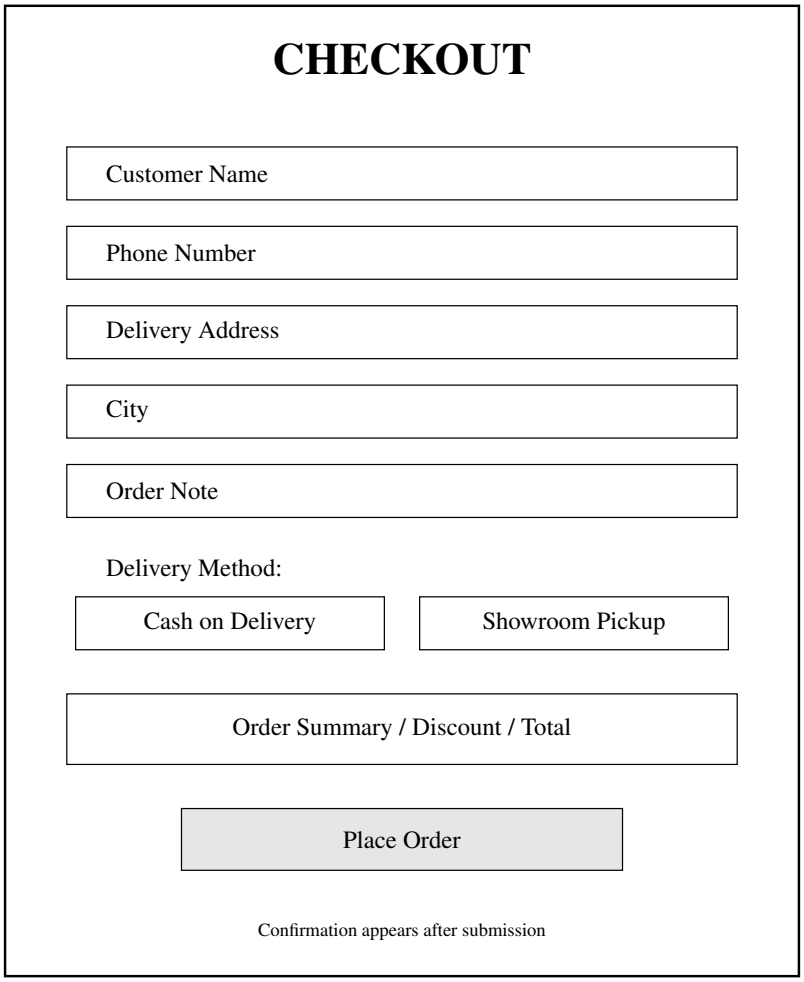


Figure 18: *Wireframe: Checkout dialog*

4.18 Reviews Section

The Reviews section displays the average rating, review count, review text and a form for a signed-in visitor to add or update a review.

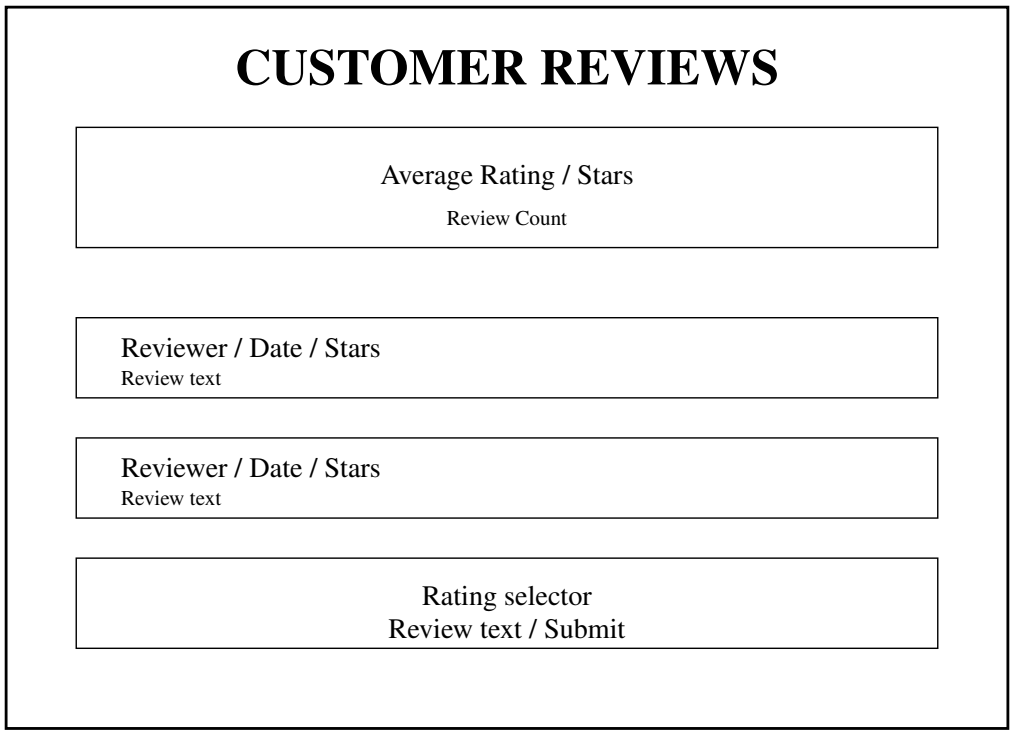


Figure 19: *Wireframe: Product reviews*

5. Design Principles

5.1 Consistency

The navigation, product cards, headings and buttons should follow a consistent design. The same card structure is used for each category.

5.2 Simple Navigation

The main menu links to website sections. On a narrow screen, the menu is collapsed to save space.

5.3 Clear Feedback

Important actions should display a message. Examples include adding a clock to the cart, saving an account, applying a coupon and submitting a form.

5.4 Error Prevention

Required fields are checked before submission. Password confirmation is checked, cart quantities are limited by stock and empty-cart messages are shown.

5.5 Responsive Layout

The desktop layout uses more columns. The mobile layout uses fewer columns and a collapsed menu so the controls fit on a smaller screen.

6. Risks and Limitations

Area	**Risk**	**What to Remember**
Product data	JSON request may fail.	Use Live Server and check the file path.
Images and map	Remote resources may be offline.	Keep fallback images and check Internet access.
Browser storage	Saved data can be edited or cleared.	Data stays in the current browser.
Account forms	Login is a demonstration.	Do not use real passwords or sensitive information.
Stock	Stock is not shared between visitors.	Each browser keeps its own demo stock changes.
Checkout	No real payment is taken.	Orders are saved locally only.
Order status	Status changes after a short time.	It is not a live delivery update.
Support	Contact form does not send emails.	Confirmation is shown on the website only.

7. Testing Checklist

ID	Test	Expected Result
T-01	Open the website through Live Server.	Home page opens.
T-02	Load the product catalogue.	Product cards appear.
T-03	Search for a known clock.	Matching clocks appear.
T-04	Search for a name with no match.	Empty-result message appears.
T-05	Change category and sort order.	Product list changes.
T-06	Move the price slider.	Price limit and product list update.
T-07	Open a product detail page.	Details for the selected ID appear.
T-08	Add a clock to the cart.	Cart count and total update.
T-09	Try to add a sold-out clock.	Item cannot be added.
T-10	Apply the TIME10 coupon.	Discount appears.
T-11	Start checkout while signed out.	Login is requested.
T-12	Register with an invalid password.	Password rules block submission.
T-13	Enter a different confirmation password.	Error message appears.
T-14	Log in with correct details.	Header displays the user's name.
T-15	Add and remove a wishlist item.	Wishlist updates.
T-16	Submit a product review.	Review appears in this browser.
T-17	Place a demo order.	Confirmation and order history appear.
T-18	Deny location permission.	Website remains usable.
T-19	Open on a narrow screen.	Navigation and cards fit the screen.
T-20	Submit the contact form.	Confirmation appears; no email is sent.

Task Sheet Review 2

Project Ref. No.: eP/Advertisement Portal Management System/01

Project Title: Alberto Clocks

Date of Preparation of Activity Plan: 13/10/2026

No.	Task	Start Date	Days	Team Member	Status
1	Plan header, navigation and Home section.	13/10/26	1	Areeba Sabri/ Fatima Raza	Complete
2	Plan product cards and category controls.	14/10/26	2	Alishah Sabri	Complete
3	Plan Technology, Store Locator and Support.	15/10/26	1	Areeba Sabri/ Fatima Raza	Complete
4	Plan Login, Sign-Up, Register and Profile screens.	16/10/26	2	Fatima Raza	Complete
5	Plan product details, cart and checkout.	18/10/26	2	Alishah Sabri/ Fatima Raza	Complete
6	Plan mobile layout and navigation.	20/10/26	1	Areeba Sabri	Complete
7	Review the sitemap and screen flow.	21/10/26	1	All team members	Complete

Task Sheet Review 3

Project Ref. No.: eP/Advertisement Portal Management System/01

Project Title: Alberto Clocks

Date of Preparation of Activity Plan: 22/10/2026

No.	Task	Start Date	Days	Team Member	Status
1	Connect product data and test product cards.	22/10/26	2	Fatima Raza	Complete
2	Test search, sort, filters and page controls.	23/10/26	1	Fatima Raza	Complete
3	Test registration and password rules.	24/10/26	1	Areeba Sabri/ Alishah Sabri	Complete
4	Test cart, coupon and checkout demonstration.	25/10/26	1	All team members	Complete
5	Test Store Locator, contact form and mobile layout.	26/10/26	1	All team members	Complete
6	Prepare User Guide and Developer's Guide.	26/10/26	2	All team members	Complete
7	Prepare wireframes and final document review.	27/10/26	1	All team members	Complete

8. Conclusion

This Design Document describes the requirements, scope, sitemap and wireframes of Alberto Clocks. The wireframes show the layout of the Home, Products, Product Details, Login, Cart, Checkout and supporting sections.

The project is designed for learning front-end website development. It demonstrates product browsing, browser-based account forms, cart actions and a checkout flow. It does not take real payment or send orders and messages to a store service.

End of Design Document